TIC 19569810

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period        7.006222 d
  Epoch (t0)    1495.7153 BTJD
  Duration      0.1200 d
  Depth         10.923 %  (109,228 ppm)
  SNR           156.2

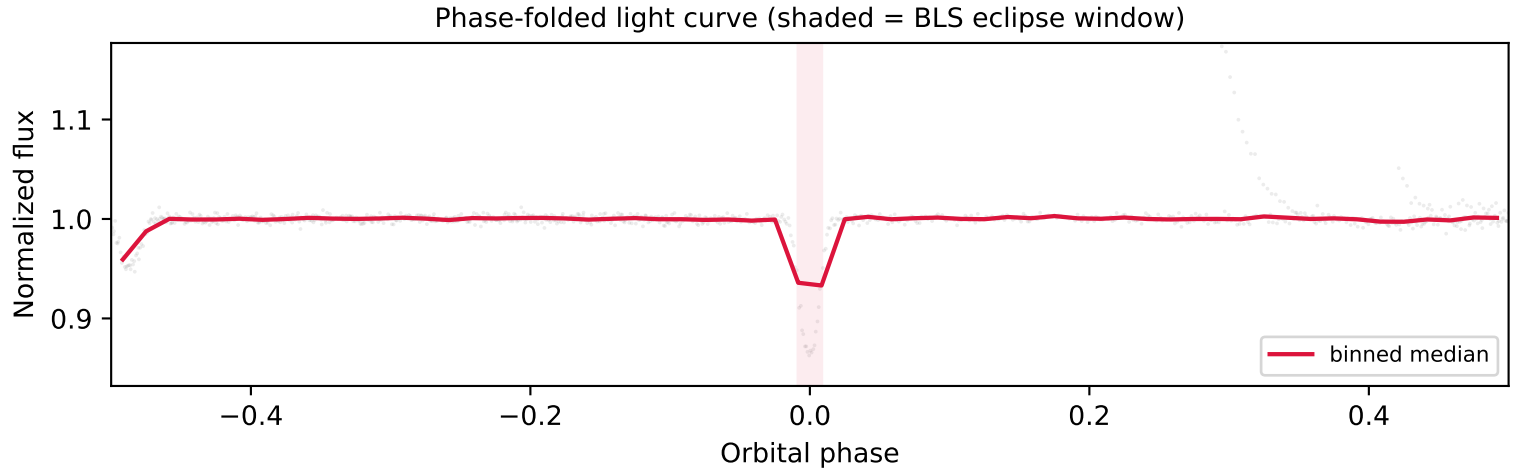
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      3  (gate >= 3)
  Robust depth ratio     1.05  (gate >= 0.50)
  Robust depth sigma     111.2  (gate >= 7.0)
  Odd/even z-score       0.46  (consistent)
  Secondary eclipse      detected, phase -0.48, 8.3 sigma
```

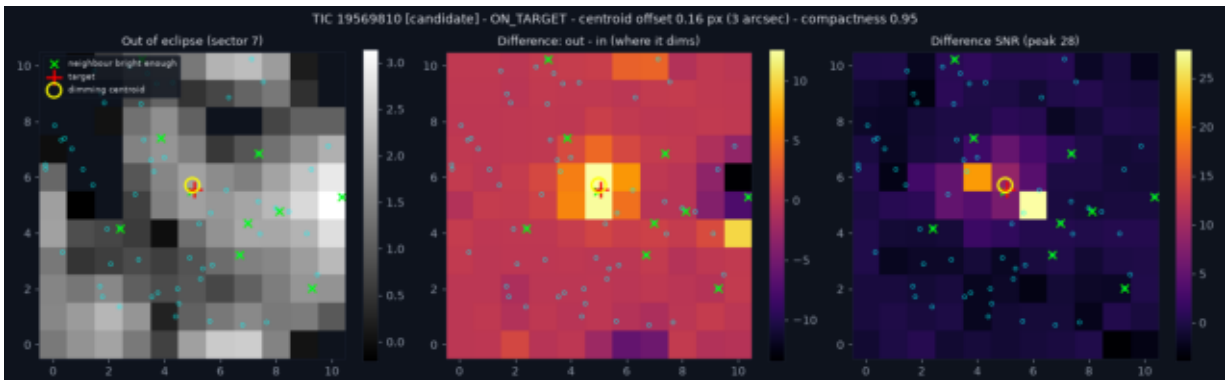
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CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 7    light-curve dip in this sector: 115.7 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   27.94   (gate >= 5.0)      Compactness   0.95   (gate >= 0.50)
  Dimming centroid offset     0.16 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 836042204
    0.22 px away, dTmag +2.38 - farther from the dimming than the target
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Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.052 px (dimming centroid 0.164 px from target vs 0.216 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.